SAT-Style Practice Test

100% Original Questions — Full-Length Exam

Section 1: Reading and Writing (Module 1 & Module 2)
Section 2: Math (Module 1 & Module 2)

Total Questions: 98

This test is designed to mirror the format, difficulty, and skill coverage of the Digital SAT. All content is original.

SECTION 1, MODULE 1: Reading and Writing

27 Questions — 32 Minutes

Question 1

Skill: Vocabulary in Context

[Easy] Tags: Vocabulary in Context

The Voyager space probes, launched in 1977, carry golden records containing sounds and images from Earth. Although originally designed for a five-year mission to study Jupiter and Saturn, the probes have far exceeded expectations. Scientist Suzanne Dodd, who manages the Voyager Interstellar Mission, continues to analyze transmissions from the probes, confirming that Voyager's contribution to our understanding of deep space is _______.

Which choice completes the text with the most logical and precise word or phrase?

A) ongoing
B) unprecedented
C) debatable
D) peripheral

Question 2

Skill: Words in Context

[Easy] Tags: Vocabulary in Context

The following text is from Khaled Hosseini's 2003 novel *The Kite Runner*. The narrator describes learning to play chess as a child.

For the tournament, I was supposed to study a book of openings my father had bought me. It was a thick, serious volume with diagrams and annotations. I was meant to memorize the Sicilian Defense and the Queen's Gambit. But I skipped ahead, glancing at the diagrams without understanding them, and then improvising, moving pieces wherever they seemed right. I never truly grasped the strategies. I daydreamed about flying kites, about being somewhere above the rooftops.

Based on the text, when the narrator describes himself as "improvising," what does he most likely mean?

A) He was violating an expectation about how to prepare for the tournament.
B) He was gaining an unfair advantage over the other competitors.
C) He was deceiving his chess instructor.
D) He was lying to himself about his strategic ability.

Question 3

Skill: Vocabulary in Context

[Easy] Tags: Vocabulary in Context

The Oxford English Dictionary's Reading Programme is a volunteer-driven initiative begun in the late nineteenth century to document the history of English words. Participants read published works and submit quotations illustrating word usage—these ________ from personal libraries and archives allow lexicographers to trace how terms such as "robot" (1920) first entered the language.

Which choice completes the text with the most logical and precise word or phrase?

A) contributions

B) resolutions

C) negotiations

D) justifications

Question 4

Skill: Vocabulary in Context

[Easy] Tags: Vocabulary in Context

Modern concert halls are typically engineered with sophisticated acoustic panels and adjustable sound systems to optimize the audience's listening experience. Because concertgoers have become accustomed to polished acoustics, performances in raw, untreated spaces can startle listeners. But bare-walled venues were likely ________ audiences centuries ago: purpose-built concert halls with advanced acoustic design were not widespread until the 1800s.

Which choice completes the text with the most logical and precise word or phrase?

A) expected by

B) confusing to

C) exciting to

D) disliked by

Question 5

Skill: Main Purpose / Main Idea

[Easy] Tags: Main Purpose / Main Idea

Struggling with an overpopulation of deer? One surprising approach may be to introduce predators back into the ecosystem. Gray wolves and mountain lions are two native predators whose reintroduction into certain U.S. regions has helped control deer herds. These apex predators are sometimes more effective at managing populations than are traditional methods such as regulated hunting.

Which choice best states the main purpose of the text?

A) To suggest an unexpected solution to a problem

B) To discuss the best way to manage wildlife

C) To describe how dangerous some predators are

D) To identify a new ecological trend

Question 6

Skill: Text Structure

[Medium]   Tags: Text Structure

Beloved, the 1987 novel by Ohio-born author Toni Morrison, is characteristic of African American literary fiction written during the 1970s and 1980s. During that era, Morrison and her contemporaries crafted haunting, psychologically complex narratives rooted in the historical trauma of slavery. Recently, however, younger Black writers have gravitated toward commercially popular genres known for being fast-paced and inventive, such as science fiction and Afrofuturism. Nigerian American author Nnedi Okofor is a prominent voice in this movement. Her 2010 novel *Who Fears Death* is a work of speculative fiction.

Which choice best describes the overall structure of the text?

A) It discusses fiction by earlier African American writers, then describes how fiction by more recent African American writers differs from the earlier fiction.

B) It praises one African American fiction writer, then criticizes a different African American fiction writer.

C) It provides an overview of African American fiction, then compares it to fiction by non-Black writers.

D) It describes one African American author's early works, then describes her more recent works.

Question 7

Skill: Function of Underlined Sentence

[Medium]   Tags: Inference / Detail

In 2016, marine biologist Danielle Dixson and colleagues published a study concluding that rising ocean temperatures significantly alter the migratory behavior of *Amphiprion ocellaris*, a species of clownfish. <u>However, Dixson's study relied on a sample of only 12 fish from a single reef system.</u> In a 2021 meta-analysis examining multiple researchers' conclusions about temperature effects on reef fish behavior, Frederick Bayer and colleagues cautioned that depending on such limited data sets can heighten the risk of skewed results. Such analysis, in turn, can lead to overstated conclusions.

Which choice best describes the function of the underlined sentence in the text?

A) It presents a criticism of the methodology reported by Dixson and colleagues in their 2016 study.

B) It lists several traits of *Amphiprion ocellaris* discovered by Dixson and colleagues while conducting their 2016 study.

C) It emphasizes a detail about where Dixson and colleagues conducted their 2016 study.

D) It states the conclusion reached by Dixson and colleagues in their 2016 study.

Question 8

Skill: Inference / Reasoning

[Medium]   Tags: Inference / Detail

The atmosphere of Venus makes direct exploration of the planet exceptionally difficult, as the dense carbon dioxide clouds and surface temperatures exceeding 450°C are known to corrode sensors and disable electronic components on landers and other exploration equipment. Using analogs, which are materials engineered to replicate the atmospheric and surface conditions of other planets, scientists are able to study characteristics of Venus's environment. Analogs like the Venus Chamber Simulant—which was developed using sulfuric acid vapor and extreme heat in laboratory settings—help researchers evaluate how well their instruments will perform under Venusian conditions.

Based on the text, what is one reason why analogs are valuable for scientists?

A) Analogs allow scientists to test the ability of research equipment to withstand some of the conditions it will encounter during a mission.

B) Scientists use analogs to track how the chemical properties of planetary atmospheres have changed over time.

C) Analogs can be combined with materials from Earth to explore how research equipment will handle extreme environments on Earth.

D) Scientists use analogs to compare the physical properties of Venus's surface to those of Earth's surface.

Question 9

Skill: Data Interpretation

[Medium] Tags: Data Interpretation

[Table: Millions of Metric Tons of Lithium Extracted in 2000 and 2022
Australia: 2000 = 5.20, 2022 = 61.00
Chile: 2000 = 5.50, 2022 = 39.00
China: 2000 = 2.80, 2022 = 19.00
Argentina: 2000 = 1.20, 2022 = 6.20]

While doing research for a paper about battery minerals, a student finds information about lithium extraction in different countries in 2000 and 2022. The student notes that Australia produced 5.20 million metric tons of lithium in 2000 and ________.

Which choice most effectively uses data from the table to complete the statement?

A) 61.00 million metric tons of lithium in 2022.

B) 6.20 million metric tons of lithium in 2022.

C) 19.00 million metric tons of lithium in 2022.

D) 39.00 million metric tons of lithium in 2022.

Question 10

Skill: Strengthen / Support Claim

[Medium] Tags: Weaken / Strengthen Hypothesis

For decades, scholars attributed the earliest known use of the zero as a placeholder to Indian mathematician Brahmagupta in 628 CE. However, historian Agathe Keller has argued that Cambodian stone inscriptions from the Khmer Empire, predating Brahmagupta by roughly a century, contain a clear numeric symbol functioning as zero. Keller asserts that in his role as an epigrapher, researcher Coedès identified <u>a dot used as a placeholder in a Khmer inscription dated to 683 CE, and these records contain the earliest datable example of the zero symbol</u>.

Which finding, if true, would most directly support the underlined claim?

A) Several dots appear in both Brahmagupta's texts and Khmer inscriptions, but the dot was commonly used in manuscripts in the 600s and 700s as a means to separate words from numbers.

B) In a Khmer stone inscription from approximately 683 CE, a dot appears between two numerals in a context that clearly indicates a positional value of nothing.

C) In a table in Brahmagupta’s astronomical text from the 620s, a small circle appears between two numbers in a context that suggests the value is not a whole number.

D) As recorded in Keller’s research, some of Coedès’s readings of Khmer inscriptions are inaccurate due to the use of whole numbers rather than placeholders.

Question 11

Skill: Data Interpretation

[Medium]   Tags: Data Interpretation

[Table: Average Monthly Sick Days per Employee at Two Points After Wellness Programs Began
Program Type | 6 Months | 18 Months
YGA (Yoga and Meditation) | 3.1 | 1.8
NTR (Nutrition Seminars) | 3.3 | 4.9]

Dr. Rachel Simmons et al. hypothesized that introducing wellness initiatives into offices would reduce absenteeism (employees missing work) and boost morale. Simmons’s team enrolled groups of corporate workers in two programs: one that provided yoga and meditation sessions (YGA) and one that offered nutrition seminars (NTR). They then measured average monthly sick days at 6 months and 18 months after the programs started. They concluded that yoga-based wellness programs were more effective at reducing absenteeism than nutrition seminars, though this result took time to become apparent.

Which choice best describes data from the table that most effectively strengthen Simmons and colleagues’ conclusion?

A) Absenteeism was largely due to low morale for the NTR group at 18 months, while it was due to illness for the YGA group at 18 months.

B) Sick days for the YGA group barely changed between 6 and 18 months, while sick days for the NTR group significantly increased during the same period.

C) Sick days were consistently higher for the NTR group over the entire measurement period, though the gap between the two narrowed over time.

D) Sick days were fairly similar for the YGA and NTR groups at 6 months, but at 18 months they had significantly increased for the NTR group and significantly decreased for the YGA group.

Question 12

Skill: Data Interpretation (Graph)

[Medium]   Tags: Data Interpretation (Graph)

[Graph: Bar chart titled “Taste Rating: Labeled vs. Unlabeled” showing average taste ratings (1=lowest, 8=highest) for six foods: “Kale Chips,” “Seaweed Snack,” “Cricket Flour Bar,” “Mealworm Pasta,” “Spirulina Smoothie,” and “Jackfruit Taco.” Each food has two bars—light gray for “unlabeled” and dark gray for “labeled.” For all six foods, the labeled bar is taller than the unlabeled bar. The gap is largest for “Cricket Flour Bar” and “Mealworm Pasta” (roughly 3-point difference) and smallest for “Kale Chips” and “Jackfruit Taco” (roughly 0.5-point difference).]

Researchers investigated how knowing what an unfamiliar food contains affects people's taste ratings. Participants rated foods once with ingredient labels visible and once without labels. For every food tested, participants who saw labels gave higher taste ratings than those who did not. But the magnitude of this labeling effect varied, as is best illustrated by the taste ratings for _______.

Which choice most effectively uses data from the graph to complete the statement?

A) "Kale Chips" and "Jackfruit Taco."

B) "Seaweed Snack" and "Mealworm Pasta."

C) "Cricket Flour Bar" and "Mealworm Pasta."

D) "Spirulina Smoothie" and "Jackfruit Taco."

Question 13

Skill: Logical Completion

[Hard] Tags: Logical Completion

Cats are overwhelmingly right-paw dominant (73–80%) and can thus be said to exhibit strong population-level right-pawedness (PLRP). Among studies of non-domestic felines, Maria K. Llorente's 2011 study of captive lions purported to show PLRP, while Craig Packer's 1986 study of wild lions did not. Overall, the studies claiming PLRP in non-domestic felines find much lower incidences of right-pawedness than domestic cats exhibit, but it's worth noting that studies of captive felines tend to show significantly greater incidences of right-pawedness than studies of wild felines do, therefore raising the possibility that _______.

Which choice most logically completes the text?

A) the number of individuals in the study of wild lions is insufficient to preclude the claim that non-domestic felines exhibit PLRP.

B) the number of individuals in the study of captive lions is insufficient for a robust claim regarding evidence of PLRP.

C) the greater exposure to humans among the captive lions induced them to acquire more right-pawed behaviors.

D) the apparent discrepancy between the studies' results may be partly attributable to the 1986 study using different criteria when assessing paw dominance than were used in the 2011 study.

Question 14

Skill: Logical Completion

[Hard] Tags: Logical Completion

The red-crowned crane and the white-naped crane are large migratory birds that winter in wetlands such as the Poyang Lake region in China. Keisuke Ueda and colleagues wanted to understand how these birds select overwintering areas. They examined habitat characteristics including water depth and the duration of seasonal flooding. They found that although only red-crowned cranes prefer areas with shallow water during winter, both red-crowned cranes and white-naped cranes prefer areas that experience seasonal flooding for more than 90 days per year. The researchers therefore concluded that neither species is very drawn to areas where _______.

Which choice most logically completes the text?

A) there is seasonal flooding for far fewer than 90 days per year.

B) species with overwintering seasons longer than 90 days are likely to be present.

C) there are any features that attract the other species.

D) there is relatively shallow water during the overwintering season.

Question 15

Skill: Logical Completion

[Hard] Tags: Logical Completion

It is challenging for publishers and printers to produce very long books that balance the demands of readability, durability, and affordability. Among some exceptionally long novels published in English, Leo Tolstoy's *War and Peace* (translated from Russian) was published in a 1,225-page edition and runs to an estimated 580,000 words, while Donna Tartt's *The Goldfinch* (written originally in English) was published in a 771-page edition and runs to an estimated 297,000 words. This indicates that _______.

Which choice most logically completes the text?

A) the edition of *The Goldfinch* has more words per page than the edition of *War and Peace*.

B) translating a text from Russian to English is likely to increase the word count.

C) page count is likely to be a more reliable indicator of how long it takes to read a book than word count.

D) the edition of *War and Peace* is printed on thinner paper than the edition of *The Goldfinch*.

Question 16

Skill: Standard English Conventions

[Medium] Tags: Verb Form / Tense

Girl with a Pearl Earring is one of the most recognized works by the Dutch Golden Age painter Johannes Vermeer. _______ around 1665, the painting can now be found at the Mauritshuis museum in The Hague.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Will complete

B) Completed

C) Completes

D) Was completed

Question 17

Skill: Standard English Conventions

[Medium] Tags: Verb Form / Tense

Engineered titanium alloy implants (TAIs) can be used to repair fractured bones and reinforce weakened joints; this capability, well suited for numerous orthopedic procedures, _______ from the exceptional strength-to-weight ratio of these alloys.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) stemming
B) having stemmed
C) stems
D) to stem

Question 18

Skill: Standard English Conventions

[Medium] Tags: Punctuation

The Sino-Tibetan language family comprises more than 400 ________ Mandarin and Cantonese, which are spoken by over 1 billion and 85 million speakers, respectively—and accounts for a significant portion of the world's linguistic diversity, making it of great interest to researchers like George van Driem.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) languages
B) languages—
C) languages:
D) languages,

Question 19

Skill: Standard English Conventions

[Medium] Tags: Punctuation

For the past 50 million years, the continents have been drifting toward their current positions at a relatively steady pace. This has not always been the ________ throughout geologic history, the continents have collided and separated multiple times through a process called the supercontinent cycle.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) case; though,
B) case, though,
C) case, though;
D) case, though

Question 20

Skill: Standard English Conventions

[Hard] Tags: Punctuation

Japanese-born visual artist Yayoi Kusama transforms gallery spaces into immersive, otherworldly environments. In her installations, mirrored walls create infinite reflections of glowing orbs, polka-dotted sculptures fill entire ________ suspended nets of light shift and shimmer overhead. This "infinity room" concept, Kusama says, "is about erasing the boundaries of the self."

Which choice completes the text so that it conforms to the conventions of Standard English?

A) rooms,

B) rooms;

C) rooms—

D) rooms:

Question 21

Skill: Transitions

[Medium] Tags: Transitions

Eighteenth-century Rococo architects championed ornamentation in their designs. Granted, the lavish gilding and curved moldings of the Amalienburg pavilion, a Rococo hunting lodge designed by François Cuvillieés, couldn't exactly pass for a natural setting. _______ one sees organic influences in Cuvillieés's preference for flowing curves over rigid geometry and his use of leaf and shell motifs.

Which choice completes the text with the most logical transition?

A) Furthermore,

B) Still,

C) Similarly,

D) In other words,

Question 22

Skill: Transitions

[Medium] Tags: Transitions

In Japan, citizens must be at least 18 years old to vote in national elections. _______ citizens in Scotland can vote in local elections at the age of 16.

Which choice completes the text with the most logical transition?

A) Firstly,

B) By comparison,

C) For example,

D) Therefore,

Question 23

Skill: Transitions

[Medium] Tags: Transitions

Generally, compact vehicles are more fuel-efficient than larger ones. For example, the aerodynamic profile of the Toyota Prius helps it achieve exceptional mileage. _______ a full-size pickup truck encounters greater air resistance, making it less fuel-efficient.

Which choice completes the text with the most logical transition?

A) Specifically,

B) On the other hand,

C) As a result,

D) In conclusion,

Question 24

Skill: Transitions

[Hard]   Tags: Transitions

As an abolitionist, Boston minister William Ellery Channing opposed slavery and advocated for human dignity and moral reform; _______ sermons he delivered at Federal Street Church in the 1830s forcefully challenged the arguments of pro-slavery advocates who defended the institution on economic grounds.

Which choice completes the text with the most logical transition?

A) in other words,

B) fittingly,

C) by comparison,

D) nevertheless,

Question 25

Skill: Rhetorical Synthesis

[Medium]   Tags: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:
• Tulum is a municipality in the state of Quintana Roo, Mexico.
• Municipalities are governmental regions responsible for providing many public services to their residents.
• One service they provide is waste management.
• Tulum covers an area of roughly 2,090 km^2.
• Quintana Roo is divided into 11 municipalities.

The student wants to emphasize the size of Tulum. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The municipality of Tulum in Quintana Roo, Mexico, covers an area of roughly 2,090 km^2.

B) Tulum—a governmental region in the state of Quintana Roo, Mexico—provides many public services to its residents.

C) Tulum is one of 11 governmental regions, known as municipalities, across Quintana Roo.

D) Providing waste management is just one example of the public services that municipalities provide.

Question 26

Skill: Rhetorical Synthesis

[Medium]   Tags: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:
• Canada has designated more than 80 areas as National Marine Conservation Areas (NMCAs).
• Some NMCAs were established specifically to protect endangered marine species.
• The Gwaii Haanas NMCA is a 3,400 km^2 area in British Columbia.
• It was established to protect endangered Pacific glass sponge reefs.
• The Fathom Five NMCA is a 112 km^2 area in Ontario.

• It was established to protect endangered freshwater shipwreck ecosystems.

The student wants to emphasize a similarity between the two NMCAs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both the Gwaii Haanas NMCA and the Fathom Five NMCA were established to protect endangered species or ecosystems.

B) Some NMCAs, such as Ontario's Fathom Five, were established specifically to protect endangered ecosystems.

C) While the Gwaii Haanas NMCA extends across 3,400 km^2, the Fathom Five NMCA encompasses only 112 km^2.

D) Canada has designated more than 80 areas as NMCAs, including the Gwaii Haanas NMCA in British Columbia.

Question 27

Skill: Rhetorical Synthesis

[Hard]   Tags: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

• Takashi Murakami (pseudonym Mr. DOB) is a Tokyo-based artist who specializes in the form of art known as Superflat.
• Blurring the boundary between commercial and fine art is a central aspect of Superflat.
• Playful mixing of traditional Japanese art with pop culture imagery is a central aspect of Superflat.
• Murakami's *Tan Tan Bo Puking* playfully mixes elements of classical Rinpa painting, manga, and corporate branding.
• Murakami: "My art is a journey through the border between high and low—a place where tradition and pop culture collide and transform each other."

The student wants to provide a specific example of playful mixing in Superflat art. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Elements of classical Rinpa painting, manga, and corporate branding are playfully mixed in the Superflat piece *Tan Tan Bo Puking* by artist Takashi Murakami.

B) The playful mixing of traditional and pop elements in Takashi Murakami's art evokes "a place where tradition and pop culture collide."

C) *Tan Tan Bo Puking* by artist Takashi Murakami is an example of Superflat, a form in which playful mixing is a central aspect.

D) Superflat art, such as Takashi Murakami's *Tan Tan Bo Puking*, blurs boundaries between commercial and fine art and playfully mixes disparate elements.

SECTION 1, MODULE 2: Reading and Writing

27 Questions — 32 Minutes

Question 1

Skill: Vocabulary in Context

[Easy] Tags: Vocabulary in Context

Like other barrier islands, the Outer Banks of North Carolina is _______: it is a constantly shifting chain of narrow sandy islands that change in width and elevation as hurricanes and ocean currents redistribute sediment along the Atlantic coastline.

Which choice completes the text with the most logical and precise word or phrase?

A) unrivaled
B) dynamic
C) sustainable
D) immutable

Question 2

Skill: Vocabulary in Context

[Easy] Tags: Vocabulary in Context

Noticing the similarity between Frida Kahlo's *Self-Portrait with Thorn Necklace* and her husband Diego Rivera's mural style, art historian Teresa del Conde argued that the former reflects Rivera's influence—that by incorporating bold colors and symbolic imagery reminiscent of Rivera's murals, Kahlo _______ a shared artistic vision. Indeed, by digitally analyzing both works, del Conde _______ the two pieces' stylistic parallels.

Which choice completes the text with the most logical and precise word or phrase?

A) corroborated
B) amalgamated
C) promulgated
D) consolidated

Question 3

Skill: Vocabulary in Context

[Medium] Tags: Vocabulary in Context

While contemporary architects may often draw inspiration from past masters like Frank Lloyd Wright, the breadth of their influences is wide. The designs of Zaha Hadid, for instance, can be said to _______ elements of both classical mathematical geometry and the organic flowing forms of the natural landscape.

Which choice completes the text with the most logical and precise word or phrase?

A) defy

B) neglect
C) foretell
D) exhibit

Question 4

Skill: Vocabulary in Context

[Hard] Tags: Vocabulary in Context

During the 2015–2016 refugee crisis, Germany accepted over one million asylum seekers from conflict zones. Political scientist Hanne Fjelde found that regional governments' responses to immigration seem to have been _______ local integration policies: the likelihood that municipalities would establish refugee support programs was 0.45 in regions with proactive outreach and 0.08 in regions with restrictive frameworks.

Which choice completes the text with the most logical and precise word or phrase?

A) predicated on
B) mediated by
C) decoupled from
D) material to

Question 5

Skill: Main Purpose

[Medium] Tags: Main Purpose / Main Idea

Though Ursula K. Le Guin, author of *The Left Hand of Darkness*, is perhaps not as widely known as the most commercially successful science fiction writers of the past half century, influential figures have championed her work, including the novelist Margaret Atwood and the literary critic Harold Bloom. In his introduction to Le Guin's novel *The Dispossessed*, Bloom praises the book's masterful synthesis of what philosopher Herbert Marcuse called the utopian and critical modes—while Marcuse believed works could be classified as either imagining ideal societies (utopian) or exposing existing flaws (critical), *The Dispossessed* demonstrates that a work can be both.

Which choice best states the main purpose of the text?

A) To present a reason why a literary critic is impressed by a certain novel
B) To explain what inspired an author to write a particular work
C) To compare the work of a writer with the work of a philosopher who admired her
D) To argue that all writing must be classified as belonging to one of two categories

Question 6

Skill: Text Structure

[Hard] Tags: Text Structure

The following text is adapted from Charles Dickens's 1853 novel *Bleak House*. Mr. Turveydrop is a dancing master in the city of London.

By the merchants of the neighborhood and its vicinity [Mr. Turveydrop] was regarded as a man of quite remarkable poise and elegance, who must present a magnificent figure in the best ballrooms of the metropolis, and by his fellow dancing masters he was esteemed as a courteous and accomplished gentleman. Mr. Turveydrop never entered into an argument with passion; he implied what others might consider, but seldom declared what he believed himself; he never permitted anyone to detect that he was performing a role, and he never gave any person cause to question his sincerity.

Which choice best describes the overall structure of the text?

A) It implies that Mr. Turveydrop's neighbors are more naïve in their estimation of him than people in London are and then explains why his neighbors have been so easily impressed.

B) It presents the favorable opinion of Mr. Turveydrop that other people hold and then describes the behaviors of Mr. Turveydrop that enable him to maintain that favorable opinion.

C) It shows that Mr. Turveydrop had originally been held in high regard by his colleagues and then details the events that caused their regard for him to subside.

D) It highlights the disparity between Mr. Turveydrop's public and private behavior and then conveys why he labors to obscure his true self from other people.

Question 7

Skill: Dual Passage Comparison

[Hard] Tags: Dual Passage Comparison

Text 1 is adapted from Virginia Woolf's 1925 novel *Mrs Dalloway*. Text 2 discusses *Mrs Dalloway*. The Strand and Bond Street are adjacent shopping streets in London through which the protagonist walks.

Text 1

To Clarissa the pavement of Bond Street had always suggested permanence. Its very stones—withdrawn a little from the roaring Strand—implied a serenity beyond the grasp of commerce. Those dignified shop fronts, unchanging, indifferent, shouldering between them a quiet clock, were fit monuments to some enduring quality, whose expression would certainly not be found in the language of shopkeeping.

Text 2

The tension between opposing states of mind in *Mrs Dalloway* is articulated through the novel's geography of London streets. On one hand, the busy Strand represents modern life's noise, speed, and commercial energy; on the other, Bond Street, accessible from the Strand, embodies an elegance that values beauty, permanence, and contemplation—what the novel treats as "the enduring."

Based on the texts, the author of Text 2 would most likely agree with which statement about Bond Street, as it is depicted in Text 1?

A) Because it is situated adjacent to the busy Strand, Bond Street symbolizes the encroachment of commercial culture into spaces the novel associates with beauty and reflection.

B) The quietness conveyed by Bond Street's appearance mirrors Clarissa's dissatisfaction with the prospect of finding authentic meaning in a world that primarily values consumption.

C) As a point of contrast with the Strand, Bond Street suggests to Clarissa the possibility of experiencing the lasting beauty promised by the contemplative spaces the novel seems to favor.

D) Bond Street has a relatively modest appearance whose sharp contrast with the more energetic atmosphere of the Strand suggests to Clarissa the dominance of commercial culture among Londoners.

Question 8

Skill: Inference / Detail

[Hard]   Tags: Inference / Detail

Brazilian architect Oscar Niemeyer's prolific career, which spanned the 1930s to the 2000s, evolved through distinct phases. After traveling to Europe in the early 1930s and immersing himself in Le Corbusier's modernist philosophy, Niemeyer began incorporating sweeping curves and open spaces derived from European modernism in his work, as seen in the Pampulha complex in Belo Horizonte, whose fluid organic forms contrast with his earlier projects in Rio de Janeiro, such as the Santos Dumont Airport, which reflect the rectilinear aesthetics of traditional Brazilian Neoclassicism.

Information in the text best supports which statement about the design of the Santos Dumont Airport?

A) It represents a transitional moment between the early and late phases of Niemeyer's development.

B) It displays the effects of Niemeyer's exposure to international architectural trends in the 1930s.

C) It is characteristic of the Rio de Janeiro architecture that influenced Niemeyer throughout his career.

D) It reflects an approach to form and shape that Niemeyer later moved away from.

Question 9

Skill: Characterization

[Hard]   Tags: Inference / Detail

The following text is from Hilary Mantel's 2009 novel *Wolf Hall*. The narrator is describing a member of the king's court.

At last, and providing a good deal of what one might call ballast, came Sir Thomas Boleyn, whose expression seemed designed to ornament a monument, or perhaps a ship's prow, and whose brow was so conspicuously furrowed with calculation that it would have been discourteous to examine too carefully the actual fruits his three decades of diplomatic service had yielded. His recently issued memoirs might have illuminated this puzzle, but since their substance did not emerge until well past the opening chapters, they might just as well have been sealed correspondence.

Based on the text, which choice best describes Sir Thomas Boleyn?

A) He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.

B) He has maintained a modest profile even though he has served the crown capably for many years.

C) He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.

D) He would be a celebrated public figure if his achievements did not have to be kept secret.

Question 10

Skill: Weaken / Strengthen Hypothesis

[Hard]   Tags: Weaken / Strengthen Hypothesis

Glasgow, Scotland, has installed green infrastructure along 72% of its riverbanks to reduce flooding and improve water quality, a practice known as riverbank naturalization. To evaluate the responses of urban bird populations to two types of naturalization—native wildflower meadows and constructed wetlands—Dr. Fiona Campbell et al. surveyed songbird communities consisting of robins, blackbirds, and 45 other species at different sites along the River Clyde. Using the Index of Songbird Community Health (ISCH), on which a high score corresponds to high community health, the researchers found that wildflower meadows are more strongly positively correlated with songbird community health than are constructed wetlands.

Which finding, if true, would most directly illustrate the researchers' finding?

A) Songbird communities at Riverside, a site with a high percentage of bank covered by meadows and wetlands, had lower average ISCH scores than did songbird communities at Parkhead, a site with a low percentage of bank covered by meadows and wetlands.

B) Songbird communities at Dalmarnock, a site with a relatively high percentage of bank consisting of meadows, had lower average ISCH scores than did songbird communities at Bridgeton, a site with a relatively high percentage of bank consisting of wetlands.

C) Songbird communities at Tollcross, a site with equal percentages of bank consisting of meadows and wetlands, had higher average ISCH scores than did songbird communities at Shettleston, a site with different percentages of meadows and wetlands.

D) The difference in average ISCH scores for songbird communities at Govan and Partick, two sites with a higher percentage of bank consisting of meadows than of wetlands, was statistically insignificant.

Question 11

Skill: Support Claim with Quotation

[Hard]   Tags: Weaken / Strengthen Hypothesis

For its 2023 exhibition *Behind the Lens*, the National Museum of Photography invited the security guards typically responsible for protecting the museum's collections to serve as curators, <u>whose individual selections from the museum's archives culminated in a show amplifying a distinct perspective not typically given a formal platform</u>.

Which quote from one of the personnel who curated the exhibition would best support the underlined claim?

A) "The most enjoyable part of curating this show has been seeing the outcome of our efforts: a unique show that spans eras, techniques, and subjects. We hope visitors will sense our affection for the pieces we are entrusted with guarding."

B) "My selections for the show were motivated by an interest in bringing attention to lesser-known photographs in the museum's collection, many of which tend to be passed over or even dismissed as unremarkable by most visitors to the museum."

C) "The photograph I chose is accompanied by a statement of the personal interpretation I've developed over countless hours standing beside the work and observing visitors' reactions—an interpretation that I've previously shared only in small impromptu conversations with those visitors who pause to discuss the piece with me."

D) “Throughout the exhibition’s development, my colleagues and I partnered closely with other museum staff whose areas of expertise lie in curation and archival preservation, and whose insights enriched our understandings of the pieces we selected for the exhibition.”

Question 12

Skill: Support Claim with Quotation

[Hard]   Tags: Weaken / Strengthen Hypothesis

A student is writing an essay on the subject of vertical farming, which involves growing crops indoors in stacked layers and is intended to help reduce the environmental impact of traditional agriculture. The student wants to make the case that vertical farming may be useful in responding to an expected rise in global food demand.

Which quotation from a publication by a researcher would most effectively support the student’s claim?

A) “Consumers tend to believe that reducing pesticide use in indoor farming would have a significant effect on food safety.”

B) “The taste of hydroponically grown produce differs across crop varieties and is also influenced by lighting conditions and nutrient solutions.”

C) “A growing population that is adding substantially more protein to its diet will contribute to an increasing demand for food production in the 21st century.”

D) “Researchers who advocate for the expansion of vertical farming claim that it is better for the environment than conventional agriculture because it requires less arable land.”

Question 13

Skill: Support / Illustrate Claim

[Hard]   Tags: Weaken / Strengthen Hypothesis

Born in Argentina in 1921, composer Astor Piazzolla was a pioneer of the *nuevo tango* (New Tango) movement that emerged in the 1950s and then spread throughout Latin America, Europe, and Japan as *nuevo tango*. Piazzolla traveled across Buenos Aires and beyond, studying classical composition techniques as well as traditional milonga rhythms, folk melodies, and other facets of Argentine musical heritage. These studies formed the foundation for the early movement’s fusion of traditional Argentine tango forms with jazz harmonies and classical structures in new compositions that represented modern urban life and pushed artistic boundaries. <u>As the movement spread beyond Argentina, the range of musical traditions woven into its fabric also expanded.</u>

Which detail about songs associated with *nuevo tango*, if true, would best illustrate the underlined claim?

A) Many feature rhythmic patterns addressing contemporary social issues that stemmed from shared experiences of political upheaval in South American countries.

B) Many demonstrate the stylistic influence of bossa nova, a genre of Brazilian music that had come to be characterized by lyrical melodies in the early 1960s.

C) Many were written with parts meant to be played on the bandoneón, a traditional instrument used across South American countries, including Argentina.

D) Many were produced by Japanese artists in the late 1970s, with others by artists in additional countries first emerging soon after.

Question 14

Skill: Logical Completion

[Hard] Tags: Logical Completion

The Clean Air Act requires US federal agencies to evaluate the air-quality effects of proposed industrial activities, such as building power plants. While many Clean Air Act assessments require public hearings, general conformity determinations (GCDs) allow streamlined reviews without hearings for activities that will have a minimal effect on air quality. In 2021 the rule governing GCDs was revised: before, GCDs could be issued for actions that "do not individually or cumulatively have a significant effect on air quality," but the revised rule allowed GCDs if actions "normally do not have a significant effect on air quality." Environmental groups found this revision to be potentially harmful because _______.

Which choice most logically completes the text?

A) the increasing need to build new industrial facilities in the US incentivizes agencies to issue GCDs after 2021 for reasons that would not have been considered valid prior to 2021.

B) the 2021 relaxation of the rule regarding GCDs would permit more streamlined reviews, resulting in more approvals by federal agencies and a paradoxically slower review process.

C) the rule that governed GCDs before 2021 allowed expedited reviews of actions that might have significant effects on air quality if those effects were believed to be rare, while the 2021 revision subjected such actions to slower reviews.

D) the 2021 revision of the rule governing GCDs would allow expedited reviews of actions that had a minimal air-quality effect when considered on their own but a significant effect when considered together.

Question 15

Skill: Logical Completion

[Hard] Tags: Logical Completion

Microbial electrolysis cells (MECs) harness the ability of certain bacteria to break down organic waste and produce hydrogen gas. The bacteria form colonies on the surface of an anode, but transferring the resulting electrons from the bacterial membrane to an external circuit requires the electrons to pass through a series of inefficient metabolic steps. Accordingly, MEC hydrogen output rarely exceeds a rate of 2.5 liters per day (L/d). In an experiment, researchers coated the anode with graphene nanoparticles. The resulting hydrogen output was 5.8 L/d. Since materials such as graphene exhibit exceptional electrical conductivity, the researchers hypothesized that _______.

Which choice most logically completes the text?

A) graphene nanoparticles may increase the metabolic processes of the bacteria, thereby increasing the number of electrons available for the circuit.

B) as the density of the bacterial colony increases, the metabolic steps may accelerate independent of the presence of the graphene nanoparticles.

C) electrons may be conducted directly to the circuit before the graphene nanoparticles catalyze the metabolic steps.

D) graphene nanoparticles may allow electrons to bypass the series of metabolic steps and transfer directly to the circuit.

Question 16

Skill: Standard English Conventions

[Medium]   Tags: Punctuation

Featured in *Radical Fiber*, a 2019 group exhibition at the Textile Museum in Washington, D.C., was the work of artist Faith Ringgold, who is best known for her narrative quilts that juxtapose painted imagery with written stories. Her work challenges conventional notions of race, gender, history, and _______ she is credited with expanding the horizons of textile-based art.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) memory,

B) memory and

C) memory

D) memory, and

Question 17

Skill: Standard English Conventions

[Medium]   Tags: Punctuation

Ratified in 2010, Kenya's constitution, according to legal _______ contains all of the eight constitutional features that promote governmental transparency. Explicit provisions for freedom of information, Odhiambo and Akech's research explains, are more likely to be found in constitutions enacted after 2000.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) researchers Odhiambo and Akech

B) researchers Odhiambo and Akech,

C) researchers, Odhiambo and Akech

D) researchers, Odhiambo and Akech,

Question 18

Skill: Standard English Conventions

[Medium]   Tags: Sentence Structure

Although both 433 Eros and 25143 Itokawa are classified as near-Earth asteroids—rocky bodies that orbit close to our planet—they exhibit striking differences in _______ object 433 Eros is considered a monolithic asteroid with a solid, unfractured interior, while 25143 Itokawa, showing a loose collection of rubble held together by gravity, is considered a rubble-pile asteroid.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) behavior. The

B) behavior; while the

C) behavior, while the

D) behavior; the

Question 19

Skill: Standard English Conventions

[Medium] Tags: Verb Form / Tense

In a parliamentary election, one might expect major parties to focus on the most densely populated districts. However, if polls and historical voting patterns suggest that the outcome in a given district is already settled, a party may choose not to campaign there. Ultimately, a district's electoral history and polling data, not its population size, _______ its strategic importance to campaigns.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) has determined

B) determines

C) determining

D) determine

Question 20

Skill: Standard English Conventions

[Hard] Tags: Verb Form / Tense

That the repeated eruptions of Mount Etna, a stratovolcano on the island of Sicily, have occurred on timescales as brief as six months—shorter intervals between eruptions than have been observed in some other volcanoes of its class—_______ to the volcano's position above an exceptionally active subduction zone and rapid magma supply from its mantle source.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is due

B) due

C) being due

D) having been due

Question 21

Skill: Standard English Conventions

[Hard] Tags: Verb Form / Tense

In her 2015 book, sociologist Arlie Russell Hochschild documents the political attitudes of working-class communities in rural Louisiana. Hochschild's ethnographic study _______ nearly 60 in-depth interviews conducted between 2011 and 2016 reveals not only persistent economic anxieties but also a deepening distrust of federal institutions, particularly those related to environmental regulation.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had examined

B) examining

C) examines

D) examined

Question 22

Skill: Transitions

[Medium] Tags: Transitions

Scientists studying asteroid deflection have focused on secondary objects such as S/2020 (2013 PY6), a moonlet orbiting the near-Earth asteroid 2013 PY6. In 2022 NASA intentionally crashed a probe into just such an object, the moonlet Dimorphos. ________ Dimorphos's orbital period around the near-Earth asteroid Didymos was permanently altered.

Which choice completes the text with the most logical transition?

A) In comparison,

B) In addition,

C) Consequently,

D) Specifically,

Question 23

Skill: Transitions

[Medium] Tags: Transitions

According to the band theory of solids, metals such as gold and aluminum do not form bonds through directional sharing of electrons. ________ metals are held together by a sea of delocalized electrons that move freely, interacting with neither specific atoms nor each other. This framework for understanding the electronic structure within metals can be used to help explain thermal and electrical conductivity.

Which choice completes the text with the most logical transition?

A) For example,

B) Rather,

C) Accordingly,

D) Likewise,

Question 24

Skill: Transitions

[Hard] Tags: Transitions

In an ecphrastic poem, a poet explores a work of visual art, such as a drawing, sculpture, or painting. John Keats's 1819 poem "Ode on a Grecian Urn," for example, meditates on an ancient Greek vase; ________ Tracy K. Smith's 2011 poem "Hubble Photographs" takes as its subject a series of deep-space images captured by the Hubble Space Telescope.

Which choice completes the text with the most logical transition?

A) nevertheless,

B) conversely,

C) likewise,

D) hence,

Question 25

Skill: Transitions

[Hard]   Tags: Transitions

Honeybee phospholipase (HbPLA) is a distinctly evolved enzyme that can cleave a substrate called phosphatidylcholine to release anti-inflammatory compounds and can also act as an antimicrobial peptide, a class of protein present in all insects. ________ HbPLA is a bifunctional enzyme whose presence indicates an insect capable of producing anti-inflammatory compounds; in contrast, the presence of antimicrobial peptides alone would be insufficient for determining this capability.

Which choice completes the text with the most logical transition?

A) Nevertheless,

B) In fact,

C) Moreover,

D) That is,

Question 26

Skill: Rhetorical Synthesis

[Medium]   Tags: Rhetorical Synthesis

A student has gathered the following information:

• The Bangladesh Garment Workers' Federation (BGWF) is a national labor union for the South Asian nation of Bangladesh.

• It advocates for improved working conditions and safety standards for garment workers.

• IndustriALL Global Union is an international federation that represents national labor unions across manufacturing sectors.

• It represents the BGWF.

Which choice most effectively uses information from the given sentences to explain the purpose of national union federations?

A) The BGWF is a national labor union in Bangladesh.

B) National labor unions, such as the BGWF, work to improve conditions for workers in their member nations.

C) The BGWF is one of the national labor unions represented by IndustriALL Global Union.

D) International federations like IndustriALL represent the national labor unions in a specific sector.

Question 27

Skill: Rhetorical Synthesis

[Hard]   Tags: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

• Generally, a material will expand when heated.

• The expansion of a material when heated is known as thermal expansion.

• A 2020 study led by Hiroshi Yamamoto and Kenji Takahashi tested the thermal expansion of various polymers.

• When a sample of polyethylene (PE) plastic was heated, its average length increased by 0.12 mm.

• When a sample of polytetrafluoroethylene (PTFE) was heated, its average length increased by 0.98 mm.

The student wants to emphasize a similarity between PE and PTFE. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2020, two research teams observed the effects of thermal expansion on various polymers, including PE and PTFE.

B) Both PE and PTFE expand when heated, according to a 2020 study.

C) Researchers determined that when heated, the average length of PTFE increased more than that of PE.

D) Heating a material will generally cause it to expand, a process known as thermal expansion.

SECTION 2, MODULE 1: Math

22 Questions — 35 Minutes

Question 1

Skill: Linear Equations

[Easy] Tags: Linear Equations

The equation **d = 25 + 3t** gives the distance d, in miles, of a cyclist from a starting point t hours after the cyclist began riding. What is the distance, in miles, of the cyclist from the starting point 5 hours after the cyclist began riding?

A) 25

B) 37

C) 40

D) 50

Question 2

Skill: Geometry (Triangles)

[Easy] Tags: Geometry (Triangles)

[Figure: A right triangle with hypotenuse labeled 25, one leg labeled 7, and the other leg labeled *a*. The right angle is at the vertex between legs 7 and *a*. Note: Figure not drawn to scale.]

Which equation shows the relationship between the side lengths of the given triangle?

A) 7a = 25

B) 7 + a = 25

C) $7^2 + a^2 = 25^2$

D) $7^2 – a^2 = 25^2$

Question 3

Skill: Data Interpretation (Graph)

[Easy] Tags: Data Interpretation (Graph)

[Figure: A histogram showing the distribution of 25 weights, in kilograms, in a data set. The first bar (0 to <10 kg) has frequency 3, second bar (10 to <20 kg) has frequency 5, third bar (20 to <30 kg) has frequency 4, fourth bar (30 to <40 kg) has frequency 6, fifth bar (40 to <50 kg) has frequency 7.]

The histogram shows the distribution of 25 weights, in kilograms, in a data set. The first bar represents the weights that are less than 10 kilograms, the second bar represents the weights that are at least 10 kilograms but less than 20 kilograms, and so on. Which of the following could be the maximum weight, in kilograms, in this data set?

A) 9

B) 29

C) 39
D) 47

Question 4

Skill: Linear Equations

[Easy] Tags: Linear Equations

If $3x + 12 = 27$, what is the value of $x + 4$?

A) 5
B) 9
C) 15
D) 19

Question 5

Skill: Linear Equations

[Easy] Tags: Linear Equations

$9x + 35 = 9x + k$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

Question 6

Skill: Linear Equations

[Easy] Tags: Linear Equations

A rope that has a length of **156 centimeters (cm)** is cut into three pieces. One piece is 52 cm long. The other two pieces have lengths that are equal to each other. What is the length, in cm, of one of the other two pieces of equal length?

Question 7

Skill: Linear Equations

[Easy] Tags: Linear Equations

A store sells notebooks for $4 each and pens for $2 each. If a customer buys a total of 15 items and spends $42, how many notebooks did the customer buy?

Question 8

Skill: Coordinate Geometry

[Medium] Tags: Coordinate Geometry

[Figure: A scatterplot showing the relationship between two variables, x and y. A line of best fit is also shown. The line passes through approximately (–12, 16) and (0, 4), with a negative slope. Points are scattered around the line.]

Which of the following equations best represents the line of best fit shown?

A) $y = x - 18.5$
B) $y = -x + 18.5$
C) $y = x - 4.0$
D) $y = -x + 4.0$

Question 9

Skill: Exponents & Polynomials

[Medium]   Tags: Exponents & Polynomials

The expression $5x^4 + 11x^4 - 9x^4$ is equivalent to bx^4, where b is a constant. What is the value of b?

Question 10

Skill: Systems of Equations

[Medium]   Tags: Systems of Equations

At what point (x, y) do the graphs of the equations $y = 3x + 5$ and $y = 4x - 3$ intersect in the xy-plane?

A) (5, 3)
B) (3, 5)
C) (8, 29)
D) (29, 8)

Question 11

Skill: Functions

[Medium]   Tags: Functions

A rectangle has a length that is 45 times its width. The function $y = (45w)(w)$ represents this situation, where y is the area, in square meters, of the rectangle and $y > 0$. Which of the following is the best interpretation of 45w in this context?

A) The length of the rectangle, in meters
B) The area of the rectangle, in square meters
C) The difference between the length and the width of the rectangle, in meters
D) The width of the rectangle, in meters

Question 12

Skill: Functions

[Medium]   Tags: Functions

$f(x) = 37$

For the given linear function f, which table gives three values of x and their corresponding values of f(x)?

A) x: 0, 1, 2 → f(x): 0, 0, 0

B) x: 0, 1, 2 → f(x): 37, 37, 37

C) x: 0, 1, 2 → f(x): 0, 37, 74

D) x: 0, 1, 2 → f(x): 37, 0, –37

Question 13

Skill: Geometry (Parallel Lines)

[Medium]   Tags: Geometry (Triangles)

[Figure: Two lines r and s are intersected by a transversal line k. At the intersection with line r, angles w° and x° are formed. At the intersection with line s, angles y° and z° are formed. Note: Figure not drawn to scale.]

In the figure shown, line k intersects lines r and s. If w = 145, which additional piece of information is sufficient to prove that lines r and s are parallel?

A) x = 35

B) y = 145

C) w + y = 180

D) y + z = 180

Question 14

Skill: Geometry (Circles)

[Hard]   Tags: Geometry (Circles)

$$x^2 - 6x + y^2 - 10y - 66 = 0$$

In the xy-plane, the graph of the given equation is a circle. If this circle is inscribed in a square, what is the perimeter of the square?

A) 20

B) 40

C) 80

D) 320

Question 15

Skill: Ratios & Percentages

[Hard]   Tags: Ratios & Percentages

A fitness studio offers a series of three bootcamp classes. 800 people attended the first class. 55% of the people who attended the first class attended the second class, and 40% of the people who attended the first and second classes attended the third class. How many people attended all three classes?

Question 16

Skill: Exponential Growth/Decay

[Hard]   Tags: Exponential Growth/Decay

$f(x) = 30(3)^{x/5}$

Which table gives four values of x and their corresponding values of f(x) for the given exponential function?

A) x: –5, 0, 5, 10 → f(x): 10, 0, 30, 60

B) x: –5, 0, 5, 10 → f(x): 10, 30, 90, 270

C) x: –5, 0, 5, 10 → f(x): –10, 30, 90, 270

D) x: –5, 0, 5, 10 → f(x): 10, 30, 60, 90

Question 17

Skill: Geometry (Area/Volume)

[Hard] Tags: Geometry (Area/Volume)

A cylindrical water tank has a radius of 5 meters and a height of 8 meters. If the tank is filled to 75% of its capacity, what is the volume of water in the tank, in cubic meters? (Use $\pi \approx 3.14$)

Question 18

Skill: Geometry (Area/Volume)

[Hard] Tags: Geometry (Area/Volume)

To study fluctuations in mineral content, samples of rock were taken from 18 quarries, each cut in the shape of a cube. The length of the edge of one of these cubes is 4.000 centimeters. This cube has a density of 0.500 grams per cubic centimeter. What is the mass of this cube, in grams?

Question 19

Skill: Quadratics / Absolute Value

[Hard] Tags: Quadratics

$-3|4x + 6| + 9 = -9$

What are all solutions to the given equation?

A) 0

B) 0 and –3

C) 0 and –14/4

D) There is no solution.

Question 20

Skill: Quadratics

[Hard] Tags: Quadratics

$x^2 - 72x - 18 = 0$

What is the sum of the solutions to the given equation?

A) 0
B) 9
C) 18
D) 72

Question 21

Skill: Geometry (Area/Volume)

[Hard]   Tags: Geometry (Area/Volume)

[Figure: A right circular cone with apex A at the top, and B is a point on the circumference of the base. The height of the cone is 12 centimeters and the slant height (segment AB, not shown) is 20 centimeters. Note: Figure not drawn to scale.]

For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is **20** centimeters. If the height of the cone is **12** centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k?

Question 22

Skill: Coordinate Geometry

[Hard]   Tags: Coordinate Geometry

In the xy-plane, line j and line m are perpendicular and intersect at the point (4, 6). If line j is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line m?

A) (5, 6 – 1/m)
B) (5, 6 + 1/m)
C) (5, 6 – m)
D) (5, 6 + m)

SECTION 2, MODULE 2: Math

22 Questions — 35 Minutes

Question 1

Skill: Functions / Graph Interpretation

[Easy] Tags: Data Interpretation (Graph)

[Graph: A parabola opening downward showing the height above ground (in meters) of a ball x seconds after the ball was launched upward from a platform. The graph passes through approximately (0, 3), peaks near (0.8, 5.2), and returns to 0 near (1.8, 0). A point is marked at (1.2, 4.5).]

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point (1.2, 4.5) in this context?

A) 1.2 seconds after being launched, the ball's height above ground is 4.5 meters.

B) 4.5 seconds after being launched, the ball's height above ground is 1.2 meters.

C) The ball was launched from an initial height of 1.2 meters with an initial velocity of 4.5 meters per second.

D) The ball reaches its maximum height of 4.5 meters at 1.2 seconds.

Question 2

Skill: Linear Equations

[Easy] Tags: Linear Equations

In one month in 2019, a freelancer earned a total of $900 by working at a primary contract and at a secondary contract doing part-time work. The equation $20h + 15c = 900$ represents this situation, where h is the number of hours worked at the primary contract and c is the number of hours worked at the secondary contract. Which of the following is the best interpretation of 20 in this context?

A) The amount, in dollars, the freelancer earned for each hour at the primary contract

B) The amount, in dollars, the freelancer earned for each hour at the secondary contract

C) The number of hours the freelancer worked at the primary contract

D) The number of hours the freelancer worked at the secondary contract

Question 3

Skill: Statistics (Standard Deviation)

[Easy] Tags: Statistics (Mean/Median)

Which of the following lists represents a data set with the smallest standard deviation?

A) 30, 31, 33, 35, 36

B) 31, 31, 33, 35, 35

C) 31, 32, 33, 34, 35

D) 32, 33, 33, 33, 34

Question 4

Skill: Geometry (Similar Triangles)

[Medium]   Tags: Geometry (Triangles)

[Figure: In the figure shown, triangle FGH is similar to triangle FIJ. The measure of angle FIJ is 62°, and FH = 18(IJ). Note: Figure not drawn to scale.]

In the figure shown, triangle FGH is similar to triangle FIJ. The measure of angle FIJ is 62°, and FH = 18(IJ). What is the measure of angle FGH?

A) 18°

B) 62°

C) (18 + 62)°

D) (18 · 62)°

Question 5

Skill: Probability

[Medium]   Tags: Probability

At a convention center, there are a total of **500** attendees. Each attendee is located in either Hall X, Hall Y, or Hall Z. If one of these attendees is selected at random, the probability of selecting an attendee in Hall X is 0.54, and the probability of selecting an attendee in Hall Y is 0.36. How many attendees are located in Hall Z?

A) 10

B) 50

C) 90

D) 180

Question 6

Skill: Systems of Equations

[Medium]   Tags: Systems of Equations

$y = x/3 + 8$
$y = -x/3 + 16$

The solution to the given system of equations is (x, y). What is the value of 2y?

A) 24

B) 20

C) 16

D) 8

Question 7

Skill: Linear Equations

[Easy] Tags: Linear Equations

In 7 days, a hummingbird consumed 31.5 grams of nectar. Which equation describes the amount of nectar y, in grams, the hummingbird consumed in these 7 days?

A) $y = 31.5 + 7$

B) $y = 31.5$

C) $y = 31.5/7$

D) $y = 7$

Question 8

Skill: Exponential Growth/Decay

[Medium] Tags: Exponential Growth/Decay

A pharmacologist observes a sample of a drug compound. An exponential model estimates that the mass, in milligrams, of the sample decreases by 18% every 8.5 minutes. Which of the following equations could represent this model, where M is the estimated mass, in milligrams, of the sample t minutes after the pharmacologist began observing?

A) $M = 50(0.18)^{t+8.5}$

B) $M = 50(0.18)^{t/8.5}$

C) $M = 50(0.82)^{t+8.5}$

D) $M = 50(0.82)^{t/8.5}$

Question 9

Skill: Coordinate Geometry

[Medium] Tags: Coordinate Geometry

Line p is defined by $y = –5x + 9$. Line q is parallel to line p in the xy-plane and passes through the point (0, 12). Which equation defines line q?

A) $y = 12x + 9$

B) $y = –12x + 9$

C) $y = 5x + 12$

D) $y = –5x + 12$

Question 10

Skill: Functions

[Hard] Tags: Functions

For the linear function g, $g(c) = –4$, where c is a constant, $g(3) = 32$, and the slope of the graph of $y = g(x)$ in the xy-plane is 6. For the linear function h, $h(c) = –5$ and $h(6) = 49$. What is the slope of the graph of $y = h(x)$ in the xy-plane?

A) –1

B) 2

C) 6

D) 8

Question 11

Skill: Exponents & Polynomials

[Hard]   Tags: Exponents & Polynomials

If $3(5 – 2x) + 8(5 – 2x) + 4 = 7(5 – 2x) + 10$, what is the value of $2x – 5$?

Question 12

Skill: Inequalities

[Hard]   Tags: Inequalities

[Graph: A coordinate plane showing a shaded region above a line. The boundary line passes through approximately (–8, –8) and (0, –6), with a positive slope of 0.25. The region above and to the left is shaded.]

The shaded region shown represents the solutions to $rx + ty \geq –48$, where r and t are constants. What is the value of $r + t$?

A) –5

B) –2

C) 2

D) 6

Question 13

Skill: Functions

[Hard]   Tags: Functions

$f(x) = \sqrt{(3x + 2)}$

The function f is defined by the given equation. If $f(a) = –3a$, where a is a constant, what is the value of a?

A) 1/3

B) 2/9

C) –2/9

D) –1/3

Question 14

Skill: Exponents & Polynomials

[Hard]   Tags: Exponents & Polynomials

$\sqrt[4]{(p^6)} = t$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{2n–1}$, where n is a constant, what is the value of n?

Question 15

Skill: Quadratics

[Hard]   Tags: Quadratics

The function g is a quadratic function. In the xy-plane, the graph of $y = g(x)$ has a vertex at (1, 4) and passes through the points (2, 38) and (–1, 140). What is the value of $g(–2) – g(0)$?

A) 108
B) 176
C) 272
D) 312

Question 16

Skill: Quadratics / Factoring

[Hard]   Tags: Quadratics

$f(x) = –2(x – 3)(x + 7)$

For the given function, what is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

Question 17

Skill: Exponential Functions

[Hard]   Tags: Exponential Growth/Decay

[Graph: The graph of y = f(x) + 3 is shown. The graph is an exponential decay curve that passes through approximately (0, 6) and approaches y = 3 as x increases. It passes through [Graph: The graph of y = f(x) + 3 is shown. The curve passes through approximately (0, 5) and decreases steeply as x increases, crossing y = 0 near x = 1.2. As x decreases toward negative infinity, the curve approaches y = 6 from below.]

The graph of $y = f(x) + 3$ is shown. Which equation defines function f?

A) $f(x) = –4^x + 1$
B) $f(x) = –4^x + 3$
C) $f(x) = –4^x + 5$
D) $f(x) = –4^x + 7$

Question 18

Skill: Ratios & Percentages

[Hard]   Tags: Ratios & Percentages

A square blueprint has a side length of 60 inches, and 1 inch on the blueprint represents an actual distance of 8 miles. A smaller version of the same blueprint is printed as a square with the side length 65% shorter than the side length of the previous blueprint. On the smaller blueprint, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 5.20

B) 12.31

C) 22.86

D) 36.57

Question 19

Skill: Quadratics

[Hard] Tags: Quadratics

$3x^2(2x - 5)(2x - u) = 0$

In the given equation, u is a positive constant. The sum of the solutions to the equation is 17/4. What is the value of u?

Question 20

Skill: Geometry (Similar Figures)

[Hard] Tags: Geometry (Area/Volume)

Right rectangular prism P is similar to right rectangular prism Q. The surface area of rectangular prism P is 72 square centimeters (cm^2), and the surface area of rectangular prism Q is 1,800 cm^2. The volume of rectangular prism Q is 2,500 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism P and right rectangular prism Q?

Question 21

Skill: Exponential Functions

[Hard] Tags: Exponential Growth/Decay

The function g is defined by $g(x) = cd^{x/m}$, where c, d, and m are constants, and d and m are integers. If g(3) = 8 and g(6) = 216, what is the value of g(9)?

Question 22

Skill: Quadratics

[Hard] Tags: Quadratics

A ball is thrown upward from a height of 6 feet. The height h, in feet, of the ball t seconds after it is thrown is given by $h = -16t^2 + 40t + 6$. How many seconds after being thrown does the ball reach its maximum height?

A) 0.75

B) 1.25

C) 2.50

D) 3.00

ANSWER KEY

Section 1, Module 1: Reading and Writing

1. A **2.** A **3.** A **4.** A **5.** A
6. A **7.** A **8.** A **9.** A **10.** B
11. D **12.** C **13.** C **14.** A **15.** A
16. B **17.** C **18.** A **19.** A **20.** B
21. B **22.** B **23.** B **24.** B **25.** A
26. A **27.** A

Section 1, Module 2: Reading and Writing

1. B **2.** A **3.** D **4.** B **5.** A
6. B **7.** C **8.** D **9.** A **10.** D
11. C **12.** C **13.** B **14.** D **15.** D
16. A **17.** D **18.** A **19.** D **20.** A
21. B **22.** C **23.** B **24.** C **25.** D
26. B **27.** B

Section 2, Module 1: Math

1. C **2.** C **3.** D **4.** B **5.** 35
6. 52 **7.** 6 **8.** D **9.** 7 **10.** C
11. A **12.** B **13.** B **14.** C **15.** 176
16. B **17.** 471 **18.** 32 **19.** B **20.** D
21. 1024 **22.** A

Section 2, Module 2: Math

1. A **2.** A **3.** D **4.** B **5.** B
6. A **7.** B **8.** D **9.** D **10.** C
11. -1.5 **12.** D **13.** D **14.** 5/4 **15.** C
16. 42 **17.** B **18.** C **19.** 7/2 **20.** 2520
21. 5832 **22.** B

Total Questions: 98
